KP05 Constructive Methods: Growing a Network from Training Points

> Level: undergraduate, year 3
>
> Pattern Recognition | Chapter 4, §4.5

Slides: read and download

Learning objectives

- Distinguish master units, ancillary units and purification, and hand-compute one purification on the four XOR points.
- Explain why the process stops using strict progress on a finite sample, and state its conditions.
- Explain why zero training error guarantees neither generalization nor a small network.

KP05 Constructive Methods: Growing a Network from Training Points

One-sentence summary: In practice we only have labelled points. A constructive method first splits them with a master unit and then handles the groups that are still mixed with ancillary units until different classes no longer share a code; this is a training-set goal, not a guarantee on unseen data.

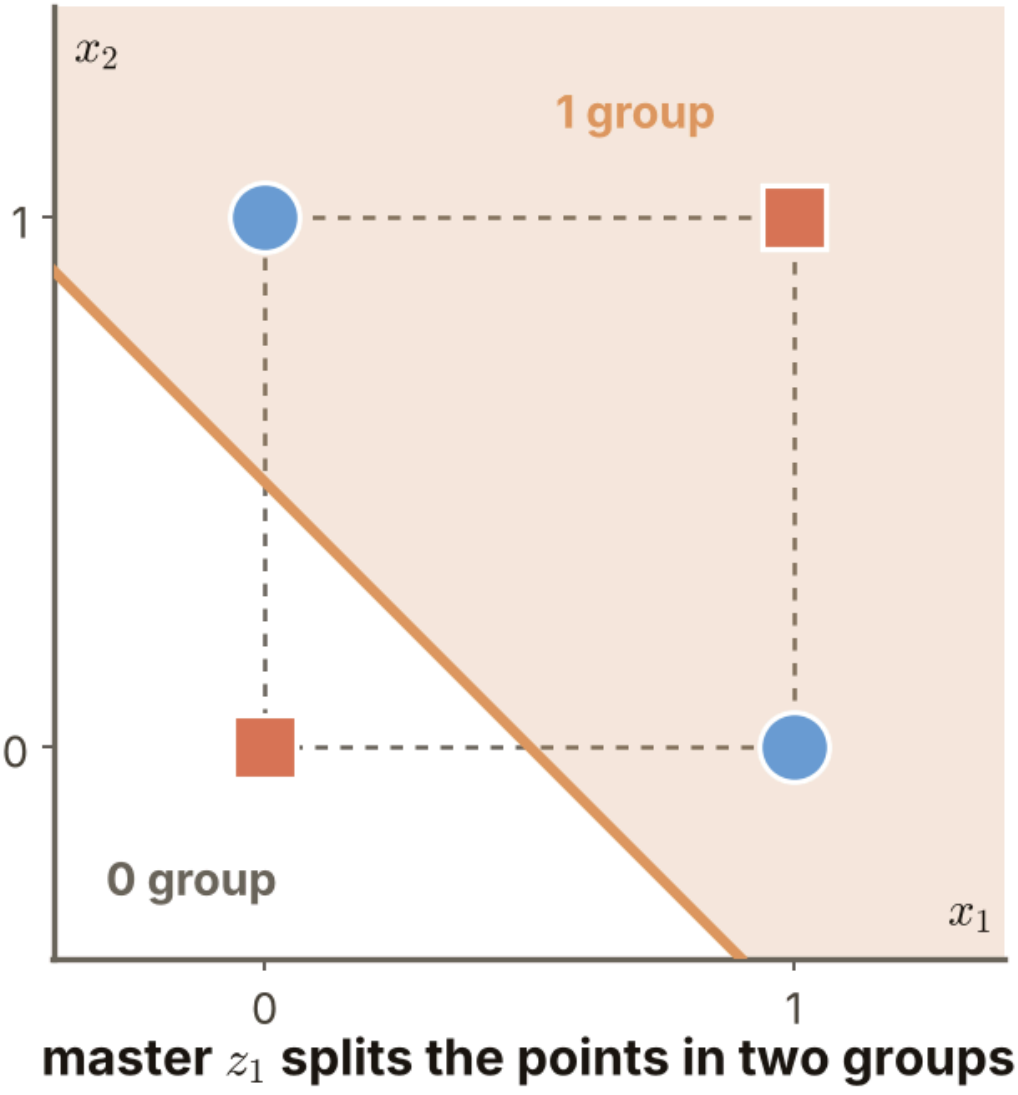
1. In practice we only have labelled points: first split them with a master unit

Plain explanation: The construction of KP04 knew the equations of the target regions, while in practice we only have points and labels. This section takes the four XOR points and starts building from points.

- **Master unit**: the first unit of a layer, in charge of classifying; it splits all points into a group with output 1 and a group with output 0.

Formal definition: let $X = \{(\mathbf{x}_n, c_n)\}_{n=1}^{N}$ be a finite labelled set with $c_n \in \{0, 1\}$. Here $z_1 = f(x_1 + x_2 - 0.5)$, where $f(t) = 1$ if $t > 0$ and 0 otherwise.

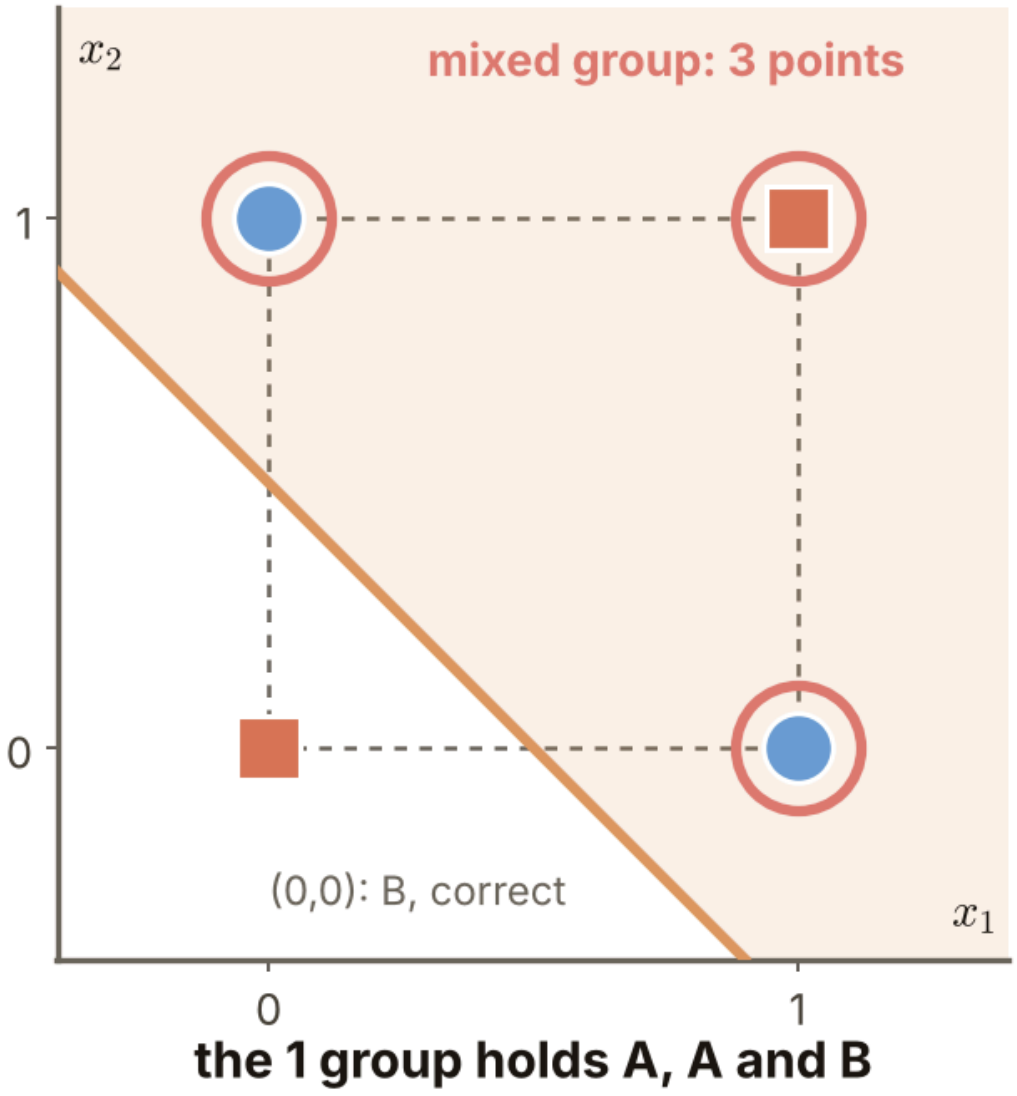
Figure `lec05-kp05-master`: above the orange line is the 1 group; only $(0, 0)$ in the lower left is in the 0 group.



2. The master gets only 3 of 4 right: its 1 group holds two A and one B

Mixed group: an output group that contains points of both classes. $(0, 0)$ alone is in the 0 group (B, correct). The 1 group is $(0, 1)$, $(1, 0)$ and $(1, 1)$, all called A: the two A are right and the B at $(1, 1)$ is wrong. The master gets 3 points right.

Figure `lec05-kp05-mixed`: red rings mark the three points of the mixed group.

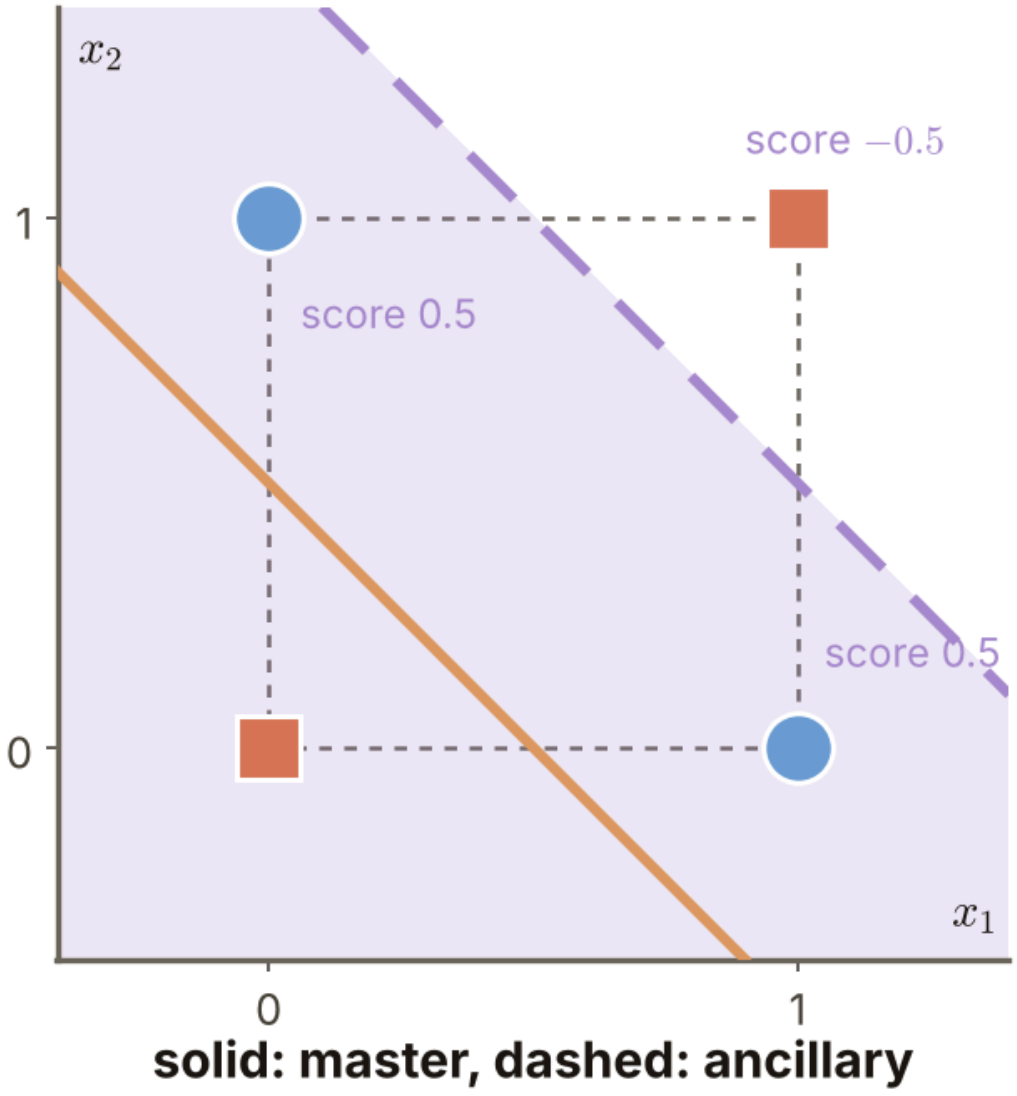


3. Add an ancillary unit trained on the three mixed points only

Ancillary unit: a new unit trained only on the points of a mixed group; it gives every point one more bit.

Example [added]: take $z_2 = f(1.5 - x_1 - x_2)$, whose line $x_1 + x_2 = 1.5$ lies exactly between $(1, 1)$ and $(0, 1), (1, 0)$: $(0, 1)$ and $(1, 0)$ score 0.5 and get bit 1; $(1, 1)$ scores -0.5 and gets bit 0. $(0, 0)$ is not in the mixed group and takes no part in training; its $z_2 = f(1.5) = 1$ is just a value computed along the way. This is a mechanism demonstration, not a recorded run of a training algorithm.

Figure `lec05-kp05-anc`: the solid line is the master unit, the violet dashed line is the ancillary unit.



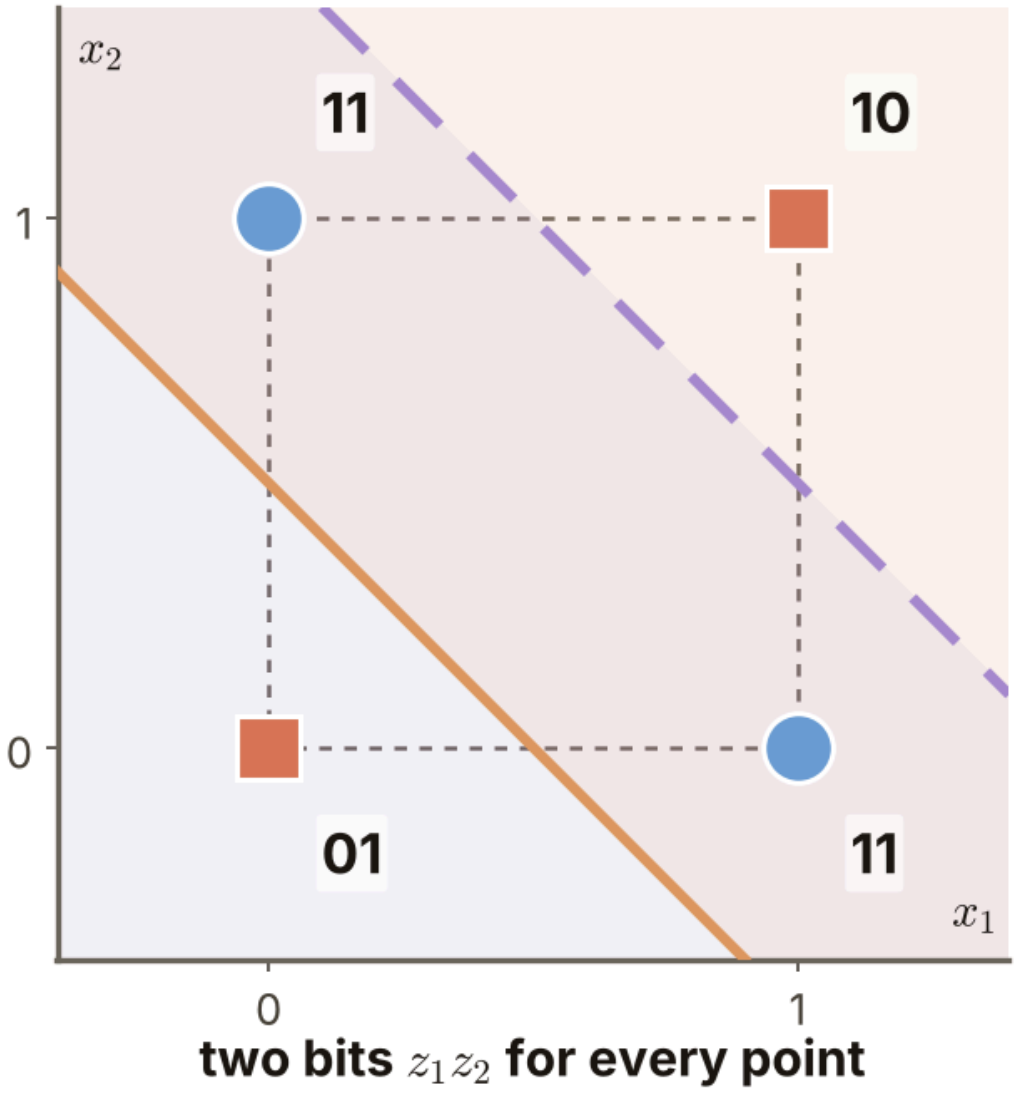
4. With the extra bit the codes are 01, 11, 11, 10

Purification: adding bits so that points of different classes no longer share exactly the same code.

Input	Code $z_1 z_2$	Label
$(0, 0)$	01	B
$(0, 1)$	11	A
$(1, 0)$	11	A
$(1, 1)$	10	B

Purification does not require one code per class: B occupying the two codes 01 and 10 is fine.

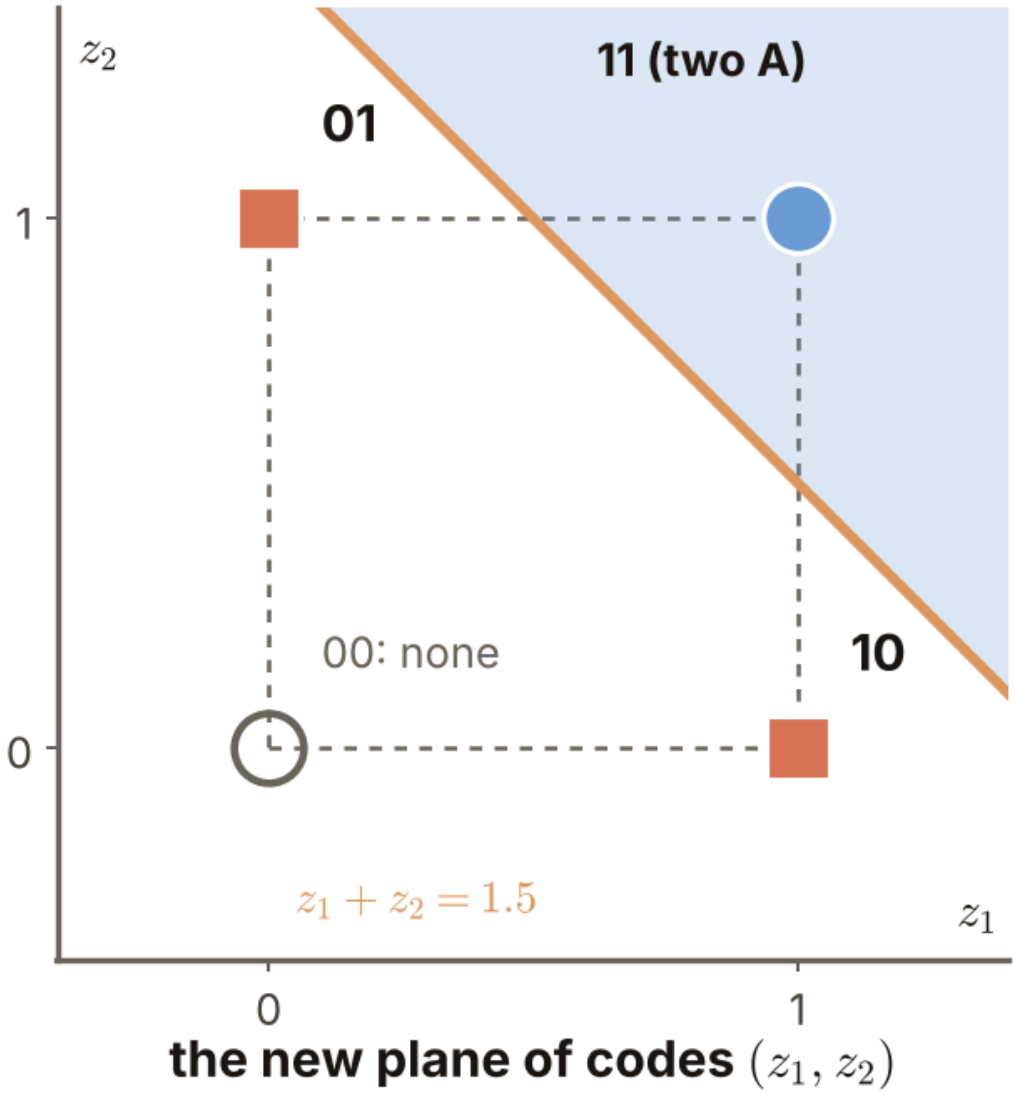
Figure `lec05-kp05-codes`: next to each point is its two-bit code.



5. On the new codes, one new master unit classifies all four points

The new master unit is $z_1 + z_2 - 1.5$, above 0 means A: 11 gives $1 + 1 - 1.5 = 0.5$, A; 01 gives -0.5 and 10 gives -0.5, B. All four points are right; merging the two A points at 11 is harmless because they are the same class.

Figure `lec05-kp05-code-plane`: the new (z_1, z_2) plane; the two A points merge at 11 and 00 has no point.



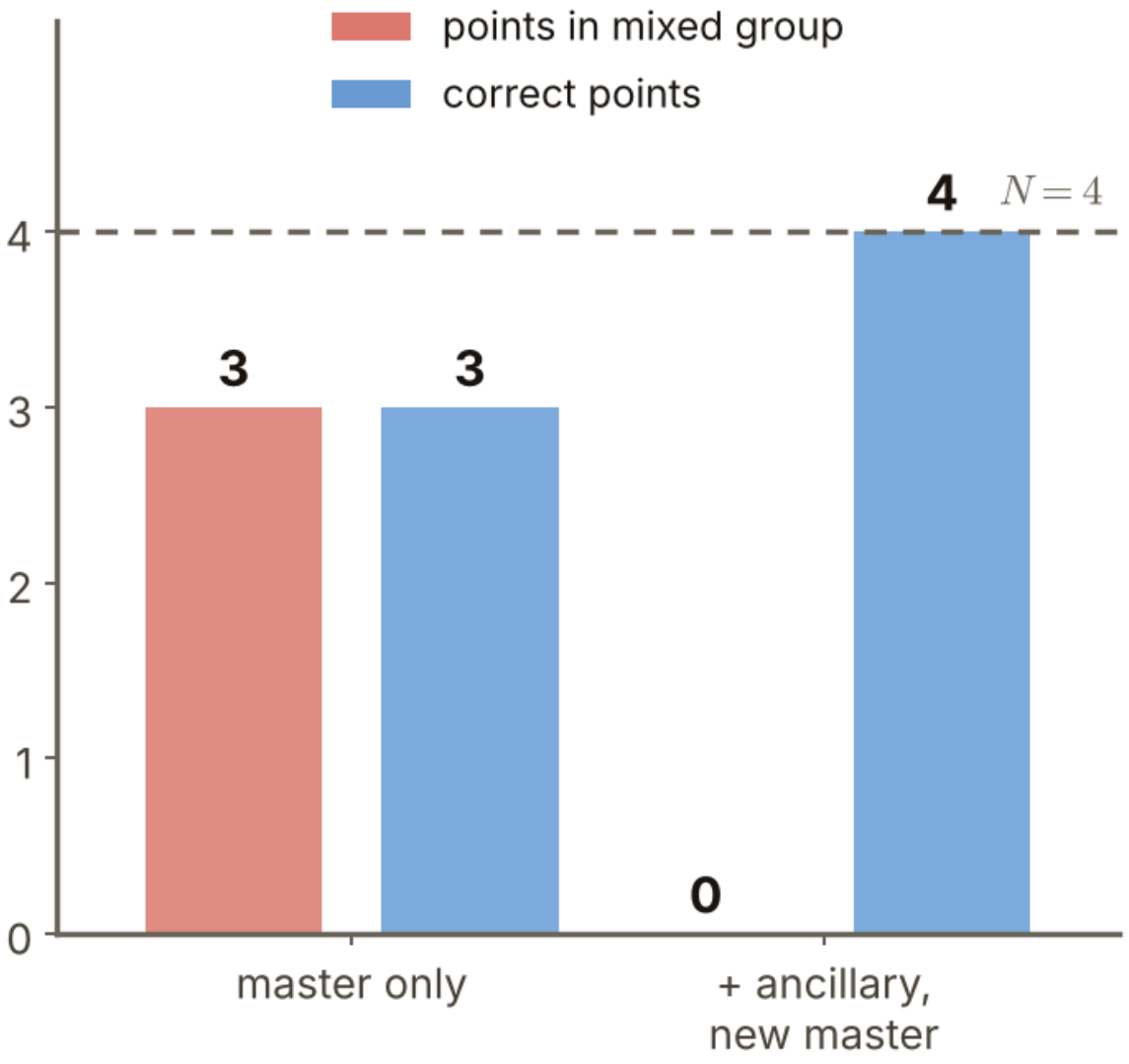
6. Why it stops: the mixed group shrinks, correct points increase, and the number of points is finite

- **Progress one**: within a layer, the mixed group that still needs refining strictly shrinks. Here it goes from 3 points to 0.
- **Progress two**: under a suitable inter-layer construction the new master keeps the points the old one got right and gets at least one more. With C_k the number of points the master of layer k gets right:

$$C_{k+1} \geq C_k + 1, \qquad C_k \leq N.$$

The construction guarantees this condition and it is not proved here; here C rises from 3 to 4. An integer cannot increase forever, so the process stops. Note that 3→0 counts points in the mixed group, not an error rate. This is the stopping reason when the conditions hold, not a claim that every nonseparable subgroup can be perfectly separated.

Figure `lec05-kp05-progress`: red is the number of points in the mixed group, blue is the number of correct points, the dashed line is $N = 4$; the values are computed for this example, not an empirical curve from the source.

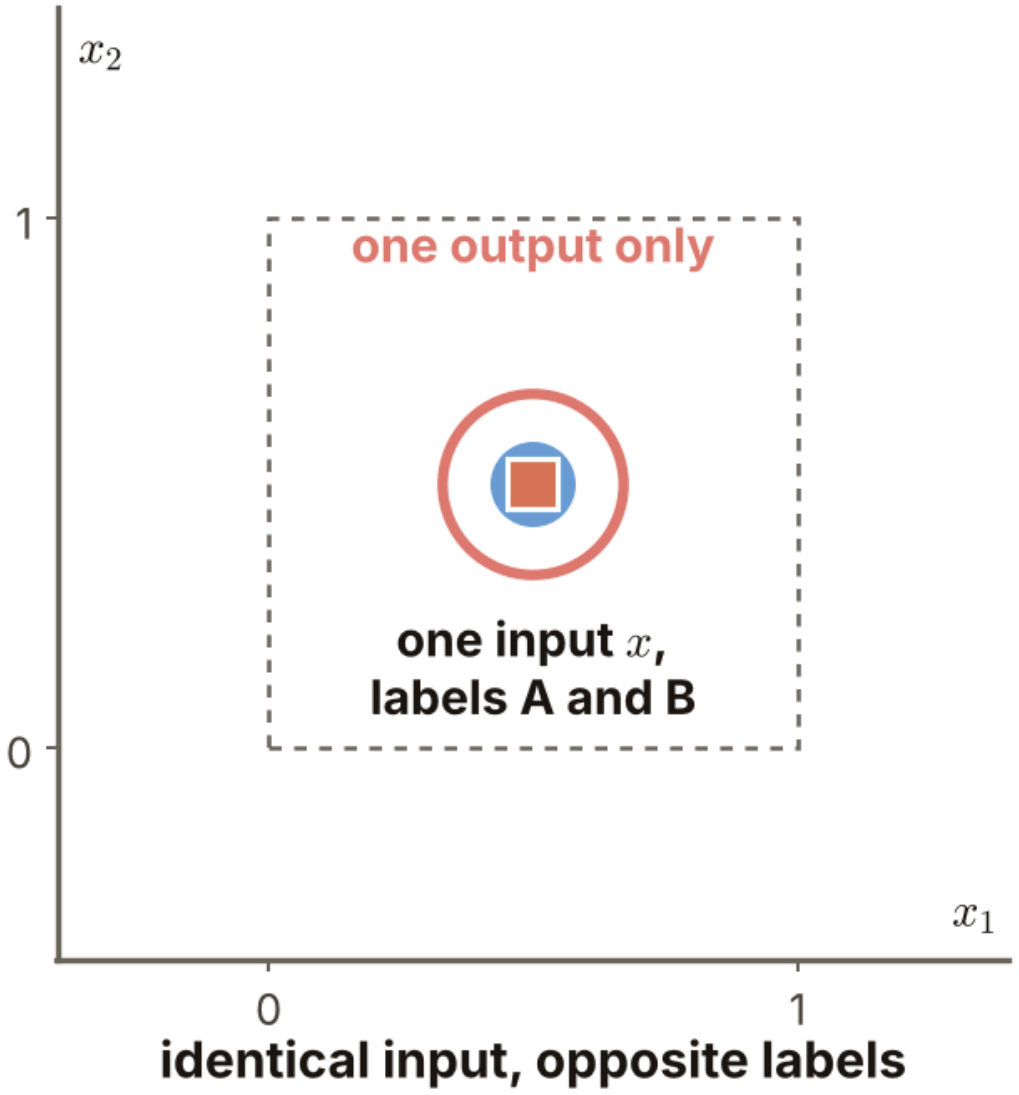


7. Stopping has conditions: labels must not contradict, and all correct only means these points

When two identical inputs carry different labels, a deterministic network has one output for one input and cannot get both right. Nodes must be added by rule to get strict progress; adding neurons arbitrarily does not guarantee termination.

Common misconception: since all training points end up right, new data will be right as well. All correct concerns only the points given; it guarantees neither correct answers at unseen locations nor a small network.

Figure `lec05-kp05-conflict`: a circle and a square on the same location with opposite labels.



The next KP asks: can we fix the architecture first and then learn the weights with a cost function?

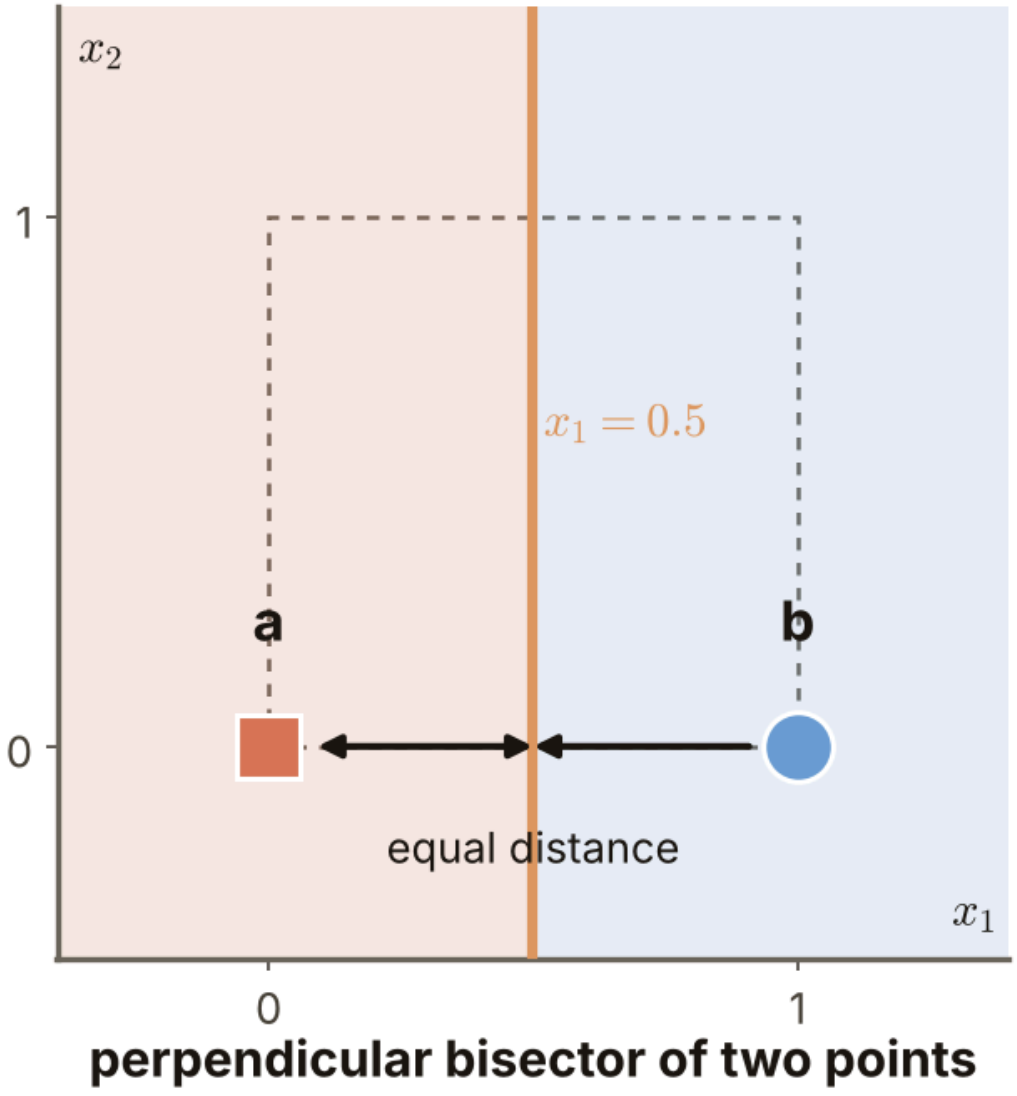
Extension: nearest neighbour, three layers, practice and worked solution

Another route (nearest neighbour): compare distances with the perpendicular bisector of two training points. [Added] Let $\mathbf{a}, \mathbf{b}$ be two training points. The equal-distance condition expands, and the common quadratic term cancels:

$$\|\mathbf{x}-\mathbf{a}\|^2 = \|\mathbf{x}-\mathbf{b}\|^2 \iff 2(\mathbf{b}-\mathbf{a})^T\mathbf{x} = \|\mathbf{b}\|^2 - \|\mathbf{a}\|^2.$$

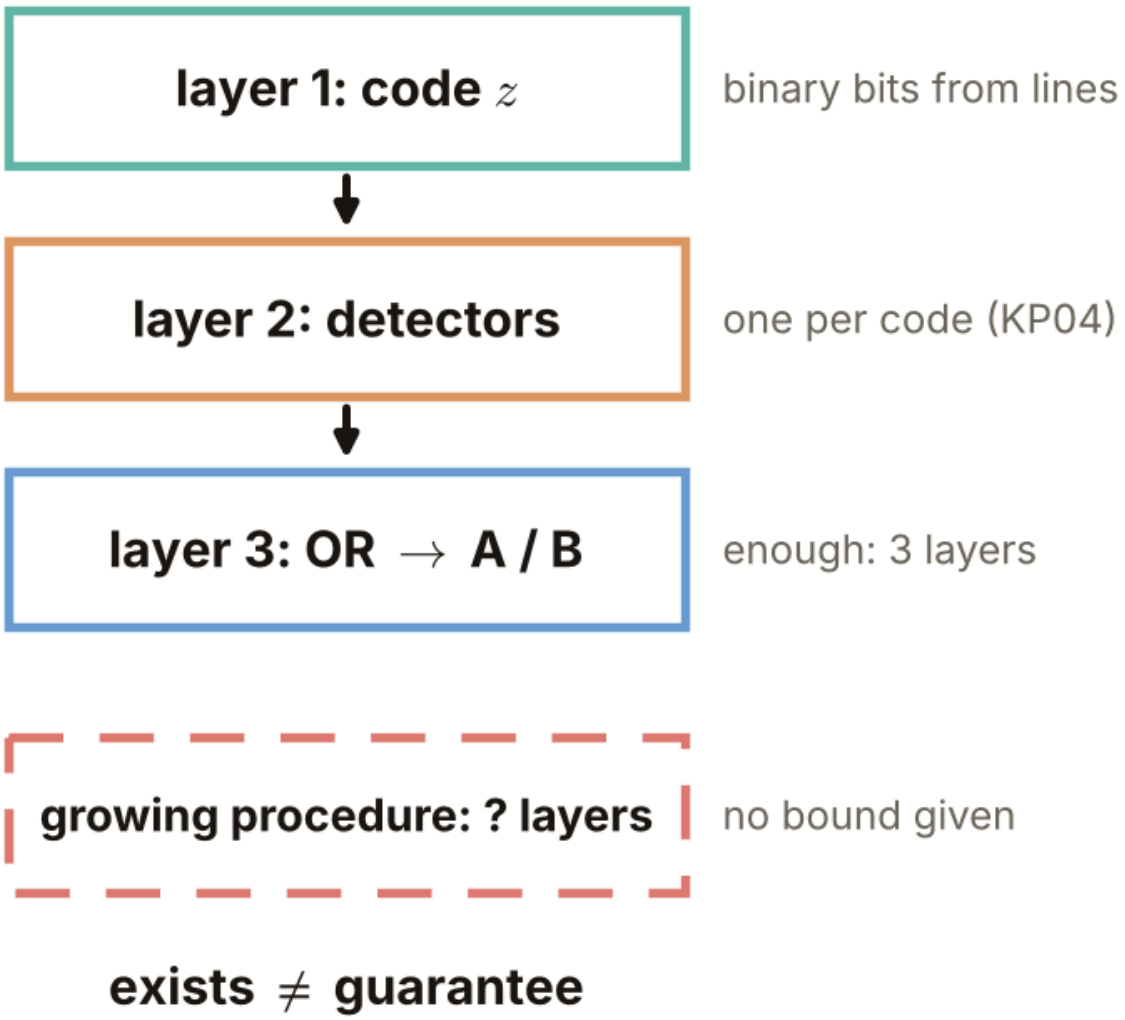
With $\mathbf{a} = (0, 0)$ and $\mathbf{b} = (1, 0)$ this gives $2x_1 = 1$, that is $x_1 = 0.5$. This is only one pairwise comparison, not a full nearest-neighbour network experiment. N points have at most $N(N-1)/2$ pairs, so comparisons can grow quadratically; this does not mean all must be kept and it is not a minimal network size.

Figure `lec05-kp05-nearest`: two training points a, b and their perpendicular bisector $x_1 = 0.5$.



Three layers suffice, which is not a bound on the growth: inputs after the first layer are binary codes, and KP04's detectors plus OR can classify the purified codes, so a suitable three-layer representation is enough; but a layer-by-layer growth procedure is not guaranteed to stop within three layers. Existence and a depth guarantee for a procedure are not the same theorem.

Figure `lec05-kp05-layers`: inputs of later layers are binary codes and detectors plus OR are enough; the growth procedure has no stated bound on the number of layers.

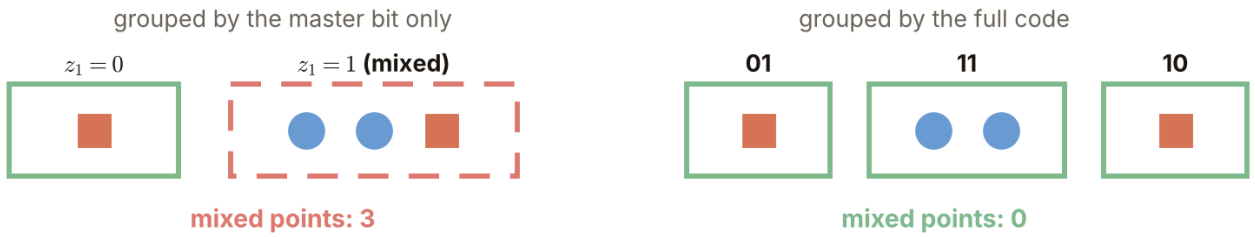


Practice:

1. Group the four points by their full code, list the labels of each group and count the points in mixed groups. Hint: first write down which points have each of the codes 01, 11 and 10.
2. If one input appears with contradictory labels, can adding more nodes classify both correctly? Hint: deterministic outputs.

Worked solution of questions 1 and 2: (1) By master bit: $\{(0,0)\}$ is B, pure; $\{(0,1),(1,0),(1,1)\}$ is A, A, B, mixed, 3 points. (2) By full code: 01: B; 11: A, A; 10: B, all pure, 0 mixed points. (3) 3→0 counts points still mixed; an error rate counts wrong answers, and the master alone is wrong on 1 point. (4) The same input gives the same output, so a deterministic network gives one answer and cannot be right for both A and B.

Figure `lec05-kp05-groups`: left, grouped by the master bit; right, grouped by the full code.



Terms and sources

English	中文
Constructive method	構造式方法
Master unit	主要節點
Ancillary unit	輔助節點
Mixed group	混雜群
Purification	純化
Perpendicular-bisector hyperplane	垂直平分超平面
Generalization	泛化

Source: Chapter 4 KP05, §4.5 (read 2026-10-24). The tiling, pocket, LMS, binary intermediate representation and nearest-neighbour comparison follow the source; the complete original algorithm was not available, so no full replication is claimed. The XOR purification numbers, the distance formula and the data-consistency condition are [added].